Thomas Bruys

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Third-year Ph.D. candidate at Georgia Tech developing trustworthy and confidence-aware AI systems for large-scale real-world applications across healthcare and supply chain applications.

EDUCATION

Georgia Institute of Technology **Aug 2022 – Expected December 2027**
Ph.D./M.S. in Electrical and Computer Engineering *Atlanta, GA*

IMT Atlantique **Sept 2020– Sept 2023**
Master's of Engineering - Major in Computer Science *Nantes, France*

TECHNICAL SKILLS

Software Development: Python, SQL, R, MATLAB, Java, Git, GitHub, Slurm, Jupyter, Claude Code, Codex, Azure
Machine Learning & Data Science: Data Analysis (NumPy, Pandas), Classical ML (Scikit-Learn), Deep Learning (PyTorch, PyTorch Lightning, TensorFlow), NLP (spaCy, NLTK), Uncertainty Quantification, Conformal Prediction
Generative AI: LLMs & Generative Models, Agents & Multi-Agent Systems (LangChain, LangGraph, Tool Use, Function Calling), LLM Inference & Serving (Hugging Face Transformers, vLLM, Ollama)

PROFESSIONAL EXPERIENCE

NSF AI Institute for Advances in Optimization | AI4OPT **Jan 2024 – Present**
Graduate Research Assistant *Atlanta, GA*

- Uncertainty Quantification and Risk Control for Large Language Models: *Jan 2025 – Present*
  * Enabled **LLM medical-consultation systems** to request additional information or commit to a diagnosis with **statistical guarantees** by developing risk-controlled stopping policies over **1,600 simulated consultations**.
  * Synthesized findings from **100+ research papers** in a survey on **Conformal Prediction for Generative AI**, covering free-form generation, reasoning, interactive systems, and multimodal models.
  * Provided **statistical guarantees on factuality risk** in LLM-generated **chest X-ray reports** by developing and evaluating **Conformal Prediction** methods.
- Conversational AI for Automated Cognitive Assessment: *Jan 2025 – Present*
  * Automated realistic **clinician–patient cognitive assessments** by developing a **multi-agent conversational AI pipeline** with **Llama 3.3** and the **OpenAI API** across conversations of **50+ questions**.
  * Scaled evaluation to **500 synthetic patient personas and corresponding conversations** through controlled persona generation in collaboration with the **Emory Department of Neurology**.
  * Achieved **over 80% scoring accuracy** on generated cognitive assessments, **exceeding previously reported clinician scoring accuracies**, with conversation quality and clinical realism validated by neurologists.
- Multimodal E-commerce Feature Extraction and Retrieval[1]: *Sep 2024 – Present*
  * Automated structured product-attribute extraction across a catalog of **100,000+ products** by developing pipelines with specialized language models including **NuExtract v1.5** for unstructured e-commerce data.
  * Delivered **multimodal product retrieval** across text and image data by developing retrieval systems leveraging **CLIP** and **Google multimodal embeddings**.
- ML-assisted Package Delivery Planning in Parcel Service Industry[1]: *Jan 2024 – Dec 2024*
  * Automated a previously **manual package-delivery planning workflow** by developing an ML framework combining **Conformal Prediction** with state-of-the-art tabular models including **TabPFN** and **TabICL**.
  * Achieved **over 90% predictive accuracy** while reducing planning decision time by approximately **10**×, enabling confidence-aware recommendations for operational decision-making.

Hopia | Hopia **Jan 2023 – June 2023**
Optimization Research Intern *Paris, France*

- **Automated hospital staff scheduling** by formulating and implementing a **Mixed-Integer Linear Program (MILP)** and complementary **heuristics** using **Gurobi**.
- Reduced staff-scheduling constraint violations by **30%** versus baseline methods, with the resulting optimization approach deployed in the application.

[1] Both projects were deployed in production by industrial partners in their respective sectors (fashion and parcel delivery).

PUBLICATIONS

- **To be submitted** : **T Bruys**, L Liu, A. Breithaupt, A. Levey, R Zandehshahvar, P Van Hentenryck, *ChatCog - A multi-agent LLM-powered framework for uncertainty-aware Cognitive Testing*
- A Moradi, **T Bruys**, R Zandehshahvar, P Van Hentenryck, *CoCoNet: Convex Conformal Network for Multi-dimensional Uncertainty Regions* [pdf] Submitted to ICLR 2027
- **T Bruys**, S Cheng, A Moradi, R Zandehshahvar, P Van Hentenryck, *Toward Reliable Generative AI with Conformal Prediction : A Survey* [pdf] Submitted to Journal of Artificial Intelligence Research (JAIR)
- **T Bruys**, R Zandehshahvar, A Hijazi, P Van Hentenryck, *Confidence-Aware Deep Learning for Load Plan Adjustments in the Parcel Service Industry* [pdf] Presented at AAAI 2025 and Informs 2025, Under revision at IEEE Access

SELECTED PROJECTS

Towards Empathetic and Emotionally Intelligent Conversational Agents | *Llama, RoBERTa, LoRA* **Fall 2024**

- Developed an **automated emotion detection system** on the ESConv dataset by fine-tuning **RoBERTa**.
- Generated **empathetic, emotionally aware dialogues** by fine-tuning **Llama 3.1 8B**.

Reproduction of a Nano GPT Model for Language Generation | *Multi-GPU training, Transformer Architecture* **Fall 2024**

- Reproduced a **nano GPT** model using PyTorch Lightning, training it on Shakespearean text to demonstrate effective language generation capabilities.

TEACHING & MENTORING

Instructor, Georgia Tech VIP Program | Vertically Integrated Projects **October 2024 - Present**

- Led an undergraduate research team in the Georgia Tech VIP Program on **Responsible AI in Healthcare**, mentoring students across experience levels and driving collaborative work.

Instructor, HerWill | AI for Equality Workshops & Datathon **October 2024 / April 2026**

- Co-organized workshops on **Foundations of Deep Learning & Advanced Text Processing with Python** for **1000+ students** from Bangladesh, Egypt, Syria, US, and more.

Instructor, Seth Bonder Camp | SBC in Computational and Data Science for Engineering **Summer 2024**

- Instructed **100+ students** for 3 in-person camps about the **Fundamentals of Programming**, **Introduction of Data Science** and **The basics of Deep Learning**.